Percolation by Example: What It Is and How Biophysics Uses It

Review notes

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Abstract

Percolation theory asks a simple question: when random local links are added to a system, at what point does a connected path first span the whole system? We introduce the idea through a sequence of small, worked examples that can be solved by hand or with a few lines of code: a one-dimensional chain, a branching tree, multivalent proteins forming a gel, a spring network that is connected but not rigid, and a population of cells relaying a signal to their neighbours. Each example illustrates one step of a practical recipe for deciding whether, and how, percolation applies to a given biological system: identify the nodes and links, choose a control parameter, estimate the threshold by counting, check it by simulation, and decide which kind of connectivity actually matters for the biology. We then survey where percolation appears in contemporary biophysics—biomolecular condensates, cytoskeletal and fibre networks, embryonic tissues, neuronal cultures, crowded membranes, excitable media and epidemics—and close with the main caveats.

1 Introduction

Imagine pouring water onto a porous stone. Whether the water reaches the bottom depends not on the average porosity alone but on whether the open pores form a *connected path* through the stone. This question, posed by Broadbent and Hammersley in 1957 [1], gave the field its name. In its simplest form one considers a lattice whose sites (or bonds) are independently "open" with probability p, and asks whether a cluster of open elements spans the system. For small p only small, isolated clusters exist; above a sharp threshold p_c a spanning cluster appears.

The rest of the article is organised around examples rather than formalism. Section 2 gives the minimum vocabulary. Section 3 works through six examples, each chosen to teach one idea that recurs in biological applications. Section 4 uses these ideas to read the current biophysics literature, and Section 5 collects a practical checklist and caveats.

2 The minimum vocabulary

Sites, bonds and continuum. In *site percolation* each vertex of a lattice is occupied with probability p and neighbouring occupied sites belong to the same cluster. In *bond percolation* all sites are present but each edge is open with probability p. In *continuum percolation* objects (cells, proteins, filaments) sit at random positions and are linked if they are close enough; the control parameter is then a density. Continuum models are usually the most natural for biology.

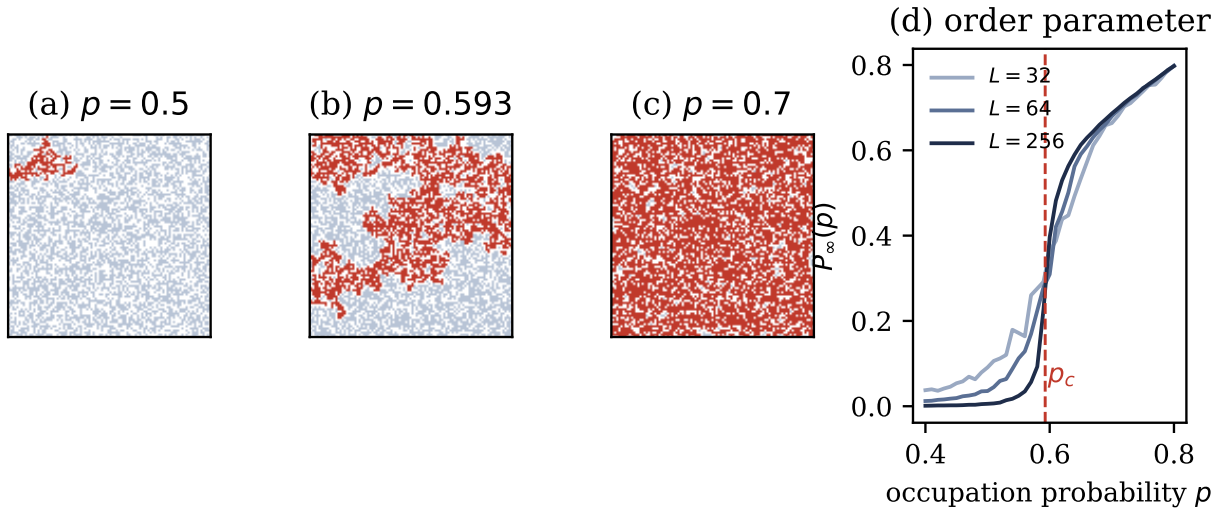


Figure 1: Site percolation on the square lattice. (a–c) Occupied sites (grey) with the largest cluster highlighted (red), below, near and above $p_c \simeq 0.593$. (d) Fraction of sites in the largest cluster, P_∞, for three system sizes L. Larger systems give a sharper onset at p_c (dashed line).

Threshold and order parameter. The *percolation threshold* p_c is the value at which a spanning cluster first appears. A convenient order parameter is P_∞, the fraction of all sites that belong to the spanning ("giant") cluster: $P_\infty = 0$ below p_c and grows above it. Below p_c the useful quantity is the *mean cluster size* χ, the average size of the cluster to which a randomly chosen occupied site belongs; it grows without bound as $p \to p_c$. Figure 1 shows both regimes on a square lattice.

Thresholds depend on details. The value of p_c depends on the lattice and on whether sites or bonds are diluted: e.g. $p_c \approx 0.593$ for site and exactly $1/2$ for bond percolation on the square lattice [4], and $p_c \approx 0.31$ (site) for the simple cubic lattice. What does *not* depend on such details is the *shape* of the transition near p_c (how P_∞ and χ behave as power laws); this is called universality [2]. For most biological questions, however, the practical issues are the location of the threshold and which control parameter moves the system across it, and those are what the examples below focus on.

Connected is not the same as rigid. A network can be connected yet mechanically floppy. *Rigidity percolation* asks when a cluster first spans the system *and* resists deformation [7, 8]. It requires more links than connectivity percolation (Example 4).

3 Worked examples: how to analyse a percolation problem

Every example follows the same three opening questions: *(i) what are the nodes and the links? (ii) what control parameter adds links? (iii) what does "spanning" mean for the biology—transport, signalling, or mechanical stability?* The examples then estimate the threshold, first by hand and then, where needed, numerically.

Example 1: A one-dimensional chain. Sites on a line are occupied with probability p. A cluster of exactly s occupied sites requires s occupied sites bounded by two empty ones, so the number of

such clusters per site is

$$n_s = (1-p)^2\, p^s. \tag{1}$$

Two different averages. Since $s\, n_s$ is the fraction of all sites that belong to clusters of size s, summing over s gives the fraction of occupied sites,

$$\sum_s s\, n_s = p. \tag{2}$$

There are now two natural ways to define a "mean cluster size":

- *Pick a cluster at random.* Every cluster counts once, whatever its size: $\langle s\rangle_{\text{cluster}} = \sum_s s\, n_s / \sum_s n_s$.
- *Pick an occupied site at random* and ask how large its cluster is. A cluster of size s is now s times more likely to be hit than a single site, so the probability that the chosen site lies in a cluster of size s is

  $$w_s = \frac{s\, n_s}{\sum_{s'} s'\, n_{s'}} = \frac{s\, n_s}{p}. \tag{3}$$

  Averaging s with this weight gives

  $$\chi = \sum_s s\, w_s = \frac{\sum_s s^2 n_s}{\sum_s s\, n_s}. \tag{4}$$

  One factor of s comes from the cluster size itself, the other from the fact that big clusters are more likely to be hit.

A small example shows the difference. Ten occupied sites split into clusters of sizes $1, 1, 1, 1, 6$. Picking a cluster gives $(1+1+1+1+6)/5 = 2$; picking a site gives $(4\times 1 + 6\times 6)/10 = 4$, because six of the ten sites sit in the big cluster. Percolation uses the site-weighted χ for two reasons: it answers the physical question "starting from one point, how many points can I reach?" (e.g. how many cells a signal started at one cell will reach), and in two and three dimensions it diverges at p_c, whereas $\langle s\rangle_{\text{cluster}}$ stays finite because it is dominated by the many small clusters.

Evaluating the sums. Using $\sum_{s\ge 1} s\, p^s = p/(1-p)^2$ and $\sum_{s\ge 1} s^2 p^s = p(1+p)/(1-p)^3$ (obtained by differentiating the geometric series $\sum_s p^s = p/(1-p)$ with respect to p and multiplying by p, once and twice),

$$\sum_s s\, n_s = (1-p)^2 \frac{p}{(1-p)^2} = p, \qquad \sum_s s^2 n_s = (1-p)^2 \frac{p(1+p)}{(1-p)^3} = \frac{p(1+p)}{1-p}, \tag{5}$$

so that

$$\chi = \frac{1+p}{1-p}. \tag{6}$$

This is finite for every $p < 1$: $\chi = 19$ at $p = 0.9$ and $\chi = 199$ at $p = 0.99$, but an infinite cluster exists only at $p = 1$. Hence $p_c = 1$.

Lesson. In one dimension a single broken link disconnects everything, so there is no transition at $p < 1$. Percolation at finite p needs alternative paths, which come from branching or from higher dimensions. The biological counterpart is a molecule with only two binding sites: it can form linear chains of any length but never a network (Example 3 with $f = 2$).

Example 2: A branching tree (Bethe lattice). Consider a tree in which every site has z neighbours and no loops. Starting from an occupied site and following one branch, the next site

is occupied with probability p and then offers $z-1$ new branches. The expected number of new occupied sites per step is therefore $p(z-1)$. If this is less than one the search dies out; if it exceeds one it can continue forever. The threshold is

$$p_c = \frac{1}{z-1}. \tag{7}$$

For $z = 3$, $p_c = 1/2$. We can do better and compute P_∞. Let Q be the probability that a given branch does *not* lead to infinity. Either its first site is empty (probability $1-p$) or it is occupied and both of its sub-branches fail:

$$Q = (1-p) + p\,Q^2 \quad \Longrightarrow \quad Q = 1 \;\text{ or }\; Q = \frac{1-p}{p}. \tag{8}$$

The key property used here is that a tree has no loops: each sub-branch is itself a copy of the original branch, statistically identical and independent of the others, so the same unknown Q appears on both sides. The quadratic $pQ^2 - Q + (1-p) = 0$ factorises as $(Q-1)\,(pQ-(1-p)) = 0$, giving the two roots. For $p < 1/2$ the second root $(1-p)/p$ exceeds one and is not a probability, so $Q = 1$: no branch ever reaches infinity. For $p > 1/2$ the physical solution is the smaller root $Q = (1-p)/p < 1$ (it is the limit obtained by iterating the equation outward from a finite depth).

A site belongs to the infinite cluster if it is occupied and at least one of its three branches succeeds, i.e. not all three fail:

$$P_\infty = p\,(1 - Q^3). \tag{9}$$

Writing $p = \frac{1}{2} + \epsilon$ gives $Q = (\frac{1}{2} - \epsilon)/(\frac{1}{2} + \epsilon) \approx 1 - 4\epsilon$, hence $Q^3 \approx 1 - 12\epsilon$ and $P_\infty \approx \frac{1}{2} \cdot 12\epsilon = 6\epsilon$: the giant cluster grows continuously from zero.

Mean cluster size by the same trick. Let T be the expected number of occupied sites that are *connected to the origin* through one given branch. Only connected sites count: as soon as the walk meets an empty site, that path is cut, and any occupied sites beyond it belong to other clusters. The first site of the branch is either

- empty (probability $1-p$): the branch contributes nothing; or
- occupied (probability p): it contributes itself (1) plus its $z-1 = 2$ outgoing sub-branches. Because there are no loops, each sub-branch is again a branch of the same tree and contributes T on average.

Hence

$$T = (1-p)\cdot 0 + p\,(1 + 2T) \quad \Longrightarrow \quad T = \frac{p}{1-2p}, \tag{10}$$

which is finite only for $p < 1/2$, recovering the threshold. (For general z, $T = p[1 + (z-1)T]$ and $T = p/[1-(z-1)p]$.) Now condition on the origin being occupied, which is what the site-weighted χ of Eq. (4) requires. Its cluster contains the origin itself plus everything reached through its three branches:

$$\chi = 1 + 3T = \frac{1+p}{1-2p}. \tag{11}$$

Note that T and χ count *sites*, not distances: because the tree keeps branching, a cluster can contain many sites while extending only a few steps from the origin. The values are $\chi = 7$, 14.5 and 74.5 at $p = 0.40$, 0.45 and 0.49 [Fig. 2(a)].

Lesson. The "branching ratio" argument, *expected number of new links per step* > 1, is the quickest way to estimate any threshold. It is exact on trees and a good first estimate elsewhere. It is also the logic behind the epidemic threshold $R_0 > 1$.

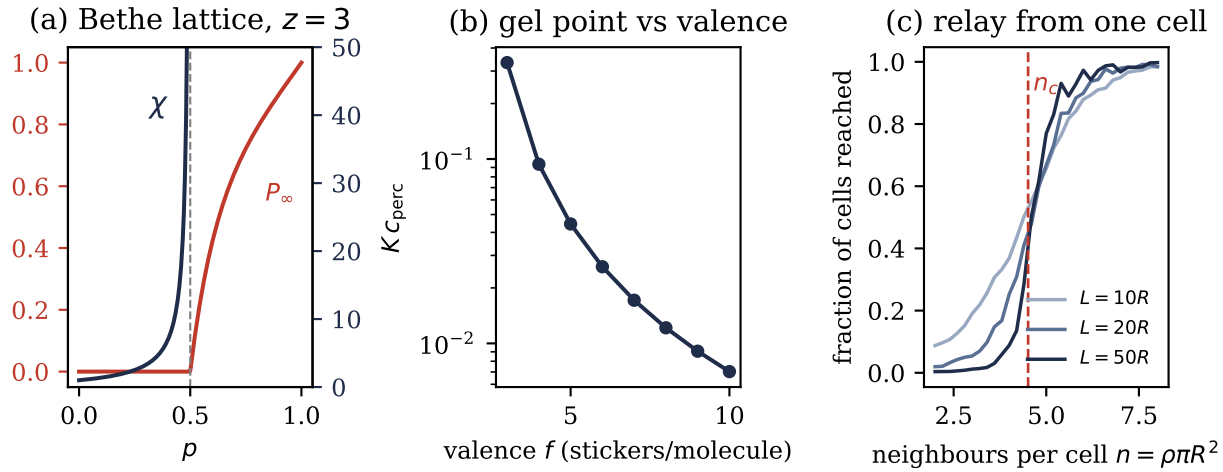


Figure 2: The worked examples. (a) Bethe lattice with $z = 3$ (Example 2): the mean cluster size χ diverges as $p \to 1/2$ from below, and the giant-cluster fraction P_∞ rises from zero above it. (b) Flory–Stockmayer gel point (Example 3): dimensionless percolation concentration Kc_{perc} versus the number of stickers per molecule; note the logarithmic axis. (c) Relay among randomly placed cells (Example 5): fraction of cells reached by a signal started at the central cell, versus the mean number of neighbours within relay range R, for three system sizes. The dashed line marks the continuum threshold $n_c \approx 4.51$.

Example 3: Multivalent proteins forming a gel. *Nodes* are protein molecules, each carrying f identical adhesive motifs ("stickers"). *Links* are reversible sticker–sticker bonds. The *control parameter* is protein concentration c (or binding affinity). *Spanning* means a network that extends through the sample: a gel.

Threshold by counting. Let α be the probability that a given sticker is bound. Arriving at a molecule through one of its bonds, that molecule has $f - 1$ other stickers, each bound (to yet another molecule, if loops are neglected) with probability α. The expected number of new molecules reached per step, the branching ratio, is therefore $\alpha(f - 1)$. This is exactly the tree of Example 2 with $z = f$ and $p = \alpha$, so by Eq. (7) the gel point is

$$\alpha_c = \frac{1}{f - 1}. \tag{12}$$

This is the classic Flory–Stockmayer result [5, 6].

Connecting to concentration. The gel condition is stated in terms of α, but experiments control the protein concentration c. Mass action connects the two. Treat each sticker as a reactant; two free stickers form one bond,

$$S + S \rightleftharpoons B, \qquad K = \frac{[B]}{[S_{\mathrm{free}}]^2}. \tag{13}$$

Express both concentrations through α:

- total sticker concentration: each molecule carries f stickers, so $[S]_{\mathrm{tot}} = fc$;
- free stickers: a fraction $1 - \alpha$ is unbound, so $[S_{\mathrm{free}}] = fc(1 - \alpha)$;
- bonds: $fc\alpha$ stickers are bound and each bond uses two of them, so $[B] = fc\alpha/2$.

Substituting into the definition of K,

$$\frac{fc\,\alpha}{2} = K\,(fc)^2(1-\alpha)^2 \quad \Longrightarrow \quad \frac{\alpha}{(1-\alpha)^2} = 2Kfc. \tag{14}$$

The left-hand side increases monotonically with α, so higher concentration or stronger binding means more stickers bound; at low concentration $\alpha \approx 2Kfc$. At the gel point $\alpha_c = 1/(f-1)$ we have $1-\alpha_c = (f-2)/(f-1)$ and

$$\frac{\alpha_c}{(1-\alpha_c)^2} = \frac{1}{f-1}\cdot\frac{(f-1)^2}{(f-2)^2} = \frac{f-1}{(f-2)^2}. \tag{15}$$

Setting this equal to $2Kfc$ gives the percolation (gel) concentration

$$c_{\mathrm{perc}} = \frac{f-1}{2Kf\,(f-2)^2}. \tag{16}$$

The derivation assumes that (i) stickers act independently and each binds at most one partner; (ii) any free sticker may pair with any other, and intramolecular bonds are not distinguished from intermolecular ones (no loops, consistent with the tree picture); (iii) the factor of 2 follows from counting identical stickers pairing with each other. Some texts include a symmetry factor $\frac{1}{2}$ in the definition of K for homotypic binding, in which case the 2 disappears; this changes c_{perc} by a constant factor but not its dependence on f. For $f = 3, 4, 5, 6$ this gives $Kc_{\mathrm{perc}} = 1/3,\ 3/32,\ 2/45,\ 5/192 \approx 0.33,\ 0.094,\ 0.044,\ 0.026$ [Fig. 2(b)]. With $K = 10^4\,\mathrm{M}^{-1}$ (a weak bond, dissociation constant $100\,\mu$M), a trivalent protein gels at $c_{\mathrm{perc}} = 2/(2\cdot 10^4\cdot 3\cdot 1)\,\mathrm{M} \approx 33\,\mu$M but a pentavalent one at about $4\,\mu$M. For $f = 2$, $c_{\mathrm{perc}} \to \infty$: chains only, no gel, as in Example 1.

Lesson. Valence is a powerful knob: adding a few binding sites lowers the gel point by an order of magnitude, even with weak individual bonds. A strong valence dependence of this kind was observed for engineered multivalent SH3-domain and proline-rich-motif proteins, whose threshold for forming droplets dropped sharply as valence increased [14]. The mean-field estimate ignores intramolecular loops (which waste bonds and raise the threshold) and ignores phase separation; combining percolation with phase separation is the subject of Section 4.1.

Example 4: Connected but floppy: a spring network. Take a two-dimensional triangular lattice in which each bond is a spring, present independently with probability p. On the full lattice every node has six bonds; each survives with probability p, so the mean number of springs per node (the *coordination number*) is $z = 6p$.

Connectivity. For connectivity we only need a path of springs. The bond-percolation threshold of the triangular lattice is known exactly, $p_c = 2\sin(\pi/18) \approx 0.347$, i.e. $z_c^{\mathrm{conn}} = 6p_c \approx 2.1$. For comparison, the tree estimate of Example 2 with $z = 6$ gives $p_c = 1/(z-1) = 0.2$; the lattice value is higher because triangles create redundant paths, a point we return to in Example 5.

Rigidity: counting degrees of freedom and constraints (Maxwell). A network is rigid if the only motions that do not stretch any spring are the trivial rigid-body motions (translations and rotations of the whole object). Count:

- *Degrees of freedom.* Each of the N nodes can move in d directions: Nd in total.
- *Trivial motions.* Translations and rotations of the whole network cost no energy: d translations plus $d(d-1)/2$ rotations, i.e. 3 in 2D and 6 in 3D.
- *Constraints.* Each spring fixes one distance and so removes one degree of freedom. Each spring is shared by two nodes, so there are $N_b = Nz/2$ springs.

If all constraints are independent, the number of *floppy modes*—non-trivial deformations that cost no energy—is

$$N_0 = Nd - \frac{Nz}{2} - \frac{d(d+1)}{2}. \tag{17}$$

Small examples. It is worth checking Eq. (17) on a few small 2D frames (Fig. 3):

- *Triangle*: 3 nodes, 6 degrees of freedom, 3 springs, 3 trivial motions: $N_0 = 6 - 3 - 3 = 0$. Rigid.
- *Square*: 4 nodes, 8 degrees of freedom, 4 springs: $N_0 = 8 - 4 - 3 = 1$. The one floppy mode is shearing the square into a rhombus without changing any side length. The square is *connected but not rigid*.
- *Square with one diagonal*: 5 springs, $N_0 = 8 - 5 - 3 = 0$. Rigid: it is two triangles sharing an edge.

Large networks. For $N \to \infty$ the trivial motions are negligible, and the floppy modes per node are

$$\frac{N_0}{N} \simeq d - \frac{z}{2}. \tag{18}$$

The network becomes rigid when this reaches zero, i.e. at the *isostatic point*

$$z \geq z_c = 2d. \tag{19}$$

In 2D, $z_c = 4$, so the condition $6p \geq 4$ gives $p_r \approx 2/3$. Careful simulations give $p_r \approx 0.660$ [8], remarkably close to the counting estimate.

Why counting is only an estimate. Eq. (17) assumes every spring removes a different degree of freedom. A spring added to a region that is already rigid (for instance a second diagonal across the square) is *redundant*: it creates internal stress but removes no floppy mode. With randomly placed springs, some regions are over-constrained while others remain floppy, so the true rigidity threshold lies slightly away from the Maxwell value. Exact algorithms (the "pebble game") count only independent constraints [8].

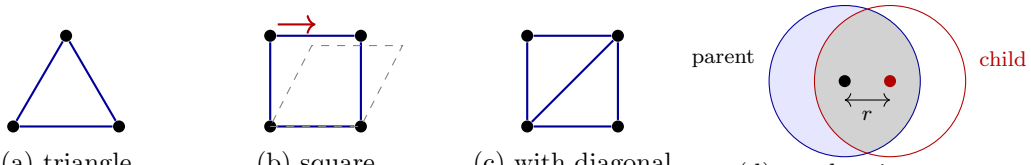
Lesson. For $0.35 < p < 0.66$ the network is *connected but floppy*, like the square in Fig. 3: pulling on one side moves the other side, yet the network offers no resistance to shear. Asking "is the network percolated?" therefore splits into two questions with two different thresholds.

Application to biopolymer networks. In 3D, Eq. (19) gives $z_c = 6$. Cytoskeletal and extracellular networks have much lower connectivity: a branch point joins three segments ($z = 3$) and a crosslink joining two crossing filaments joins four ($z = 4$). With $z = 4$, each node has 3 degrees of freedom but only $z/2 = 2$ constraints, leaving about one floppy mode per node. Such networks are nevertheless rigid because filaments also resist *bending*, which adds constraints not included in the spring counting. Below the isostatic point the shear modulus is then controlled by the (weak) bending stiffness rather than the (strong) stretching stiffness, and applied strain can push the network across a rigidity transition, producing the strong strain-stiffening typical of collagen and fibrin gels [9, 10].

Example 5: Cells relaying a signal. Cells are scattered at random in the plane with area density ρ (a Poisson point process). An excited cell excites every cell within distance R, which then relays further. A signal started at one cell therefore reaches exactly the cluster containing that cell.

Control parameter. The number of cells inside a disc of radius R around a given cell is Poisson-distributed with mean

$$n = \rho\, \pi R^2, \tag{20}$$



(a) triangle $N_0 = 6-3-3 = 0$ (b) square $N_0 = 8-4-3 = 1$ (c) with diagonal $N_0 = 8-5-3 = 0$ (d) overlapping ranges

Figure 3: (a–c) Maxwell counting on small two-dimensional spring frames (Example 4): N_0 = degrees of freedom − springs − rigid-body motions. The square (b) is connected but has one floppy mode, shearing into a rhombus (dashed). (d) In the relay problem (Example 5), a cell excited by its parent at distance $r < R$ can only recruit new cells from the part of its range (red circle) that does not overlap the parent's range (shaded overlap).

the density times the area of the disc. This mean number of neighbours is the natural control parameter.

Estimate 1: tree approximation. If every neighbour of an excited cell were new, each cell would pass the signal to n new cells, and by the branching argument of Example 2 the signal would spread when $n > 1$. This is the threshold for a random graph without geometry.

Estimate 2: subtracting the overlap with the parent. In real space, a cell that has just been excited sits within distance $r < R$ of the cell that excited it, so their ranges overlap [Fig. 3(d)]. Every cell in the overlap has already been excited by the parent and is not new. Two discs of radius R with centres a distance r apart overlap in an area

$$A(r) = 2R^2 \arccos\!\left(\frac{r}{2R}\right) - \frac{r}{2}\sqrt{4R^2 - r^2}. \tag{21}$$

Because cells are uniformly distributed, the child sits at distance r from its parent with probability density $2r/R^2$ on $0 < r < R$. Averaging,

$$\frac{\langle A \rangle}{\pi R^2} = \frac{1}{\pi R^2}\int_0^R A(r)\,\frac{2r}{R^2}\,dr = 1 - \frac{3\sqrt{3}}{4\pi} \approx 0.587. \tag{22}$$

Only about 41% of a child's range is new territory, so the branching ratio is roughly $0.41\,n$ and the threshold rises to $n_c \approx 1/0.41 \approx 2.4$.

Estimate 3: the exact answer. Estimate 2 still ignores overlaps with grandparents and siblings and the many closed loops of the neighbour graph; all of these make links redundant. The exact threshold comes from simulations of overlapping discs. The literature usually quotes it as a critical *filling factor* $\eta_c = \rho\pi a^2 \approx 1.128$ for discs of radius a that are connected when they overlap, i.e. when their centres are closer than $2a$ [15]. Our relay range is $R = 2a$, so $n = \rho\pi R^2 = 4\eta$ and

$$n_c = 4\eta_c \approx 4.51. \tag{23}$$

The successive estimates $n_c \approx 1 \to 2.4 \to 4.51$ show how strongly redundant links raise the threshold in two dimensions. The simulation in Fig. 2(c) (generated with the method of Example 6) shows the fraction of cells reached from the central cell, and how the transition sharpens as the population grows.

Numbers. For $R = 100\,\mu$m, $\pi R^2 \approx 3.1 \times 10^4\,\mu\text{m}^2$, so $\rho_c = n_c/(\pi R^2) \approx 1.4 \times 10^{-4}\,\mu\text{m}^{-2} \approx 140$ cells/mm^2 (using $1\,\text{mm}^2 = 10^6\,\mu\text{m}^2$).

Unresponsive cells. Suppose each cell can relay only with probability q (e.g. heterogeneous excitability), independently of the others, and unresponsive cells neither receive nor pass on the

signal. Removing points independently from a Poisson process leaves a Poisson process of density $q\rho$, so the responsive cells see an effective neighbour number qn. Relay spans the population only if $qn > n_c$, i.e.

$$q > q_c = \frac{n_c}{n}. \tag{24}$$

A population with $n = 9$ has $q_c \approx 0.5$: it tolerates about half of its cells being unresponsive.

Lesson. Tree-like estimates underestimate thresholds in low dimensions because of redundant, loop-forming links; geometry matters. The example also shows the limits of the picture: in real excitable systems, such as cAMP relay in *Dictyostelium* or cardiac tissue, signals diffuse and sum over many cells, so a binary "within range or not" rule is only a first approximation. The density-dependent onset of collective cAMP signalling is usually analysed as a dynamical transition [22]; percolation is most informative when coupling is genuinely short-ranged, as for gap-junction-coupled cardiac cells, where reentry was found to be most likely near the percolation threshold of excitable cells [21].

Example 6: Doing it numerically. When geometry or correlations make counting unreliable, the generic procedure is: generate a configuration, label its clusters, measure the largest cluster, repeat for many configurations and several system sizes, and look for the value of the control parameter at which the curves for different sizes sharpen and cross. For a lattice this takes a few lines of Python:

```
import numpy as np
from scipy.ndimage import label
def largest_cluster_fraction(L, p, rng=np.random.default_rng()):
    occupied = rng.random((L, L)) < p        # site percolation
    labels, n = label(occupied)              # cluster labelling
    if n == 0: return 0.0
    sizes = np.bincount(labels.ravel())[1:]  # drop background
    return sizes.max() / L**2
```

For off-lattice systems (cells, proteins, filaments) one builds a neighbour graph, e.g. with a k-d tree, and labels connected components with a graph library; Fig. 2(c) was produced that way. The same code applied to segmented microscopy images (cells as nodes, measured contacts as links) turns the theory into an analysis tool for data.

4 Is percolation discussed in contemporary biophysics?

Yes, at every scale from proteins to populations. Table 1 lists the main areas and the example above that captures each one.

4.1 Biomolecular condensates

The most active current use of percolation is in the physics of biomolecular condensates—membraneless compartments such as nucleoli and stress granules. Condensate-forming proteins and RNAs are treated as *associative polymers* of "stickers" and "spacers" [11]: precisely the setting of Example 3. In the widely used framework of *phase separation coupled to percolation* (PSCP), a density transition (phase separation, threshold c_{sat}) and a networking transition (percolation, threshold c_{perc}) act together [12]. Phase separation concentrates molecules into droplets; if the dense-phase concentration exceeds c_{perc}, the droplet interior is a connected network (Fig. 4). Condensates are then viscoelastic

Table 1: Where percolation appears in biophysics.

System	Nodes / links	Control parameter	Kind	Example
Condensates	proteins, RNA / sticker bonds	concentration, valence	connectivity (gel)	3
Cytoskeleton, ECM	filaments / crosslinks	crosslinker density	connectivity and rigidity	4
Embryonic tissue	cells / adhesive contacts	contacts per cell	rigidity	4
Neuronal cultures	neurons / synapses	synaptic strength	bond (graph)	2
Membranes	free area / obstacles	obstacle area fraction	site; diffusion	6
Excitable tissue, epidemics	cells, hosts / transmission	coupling, R_0	bond	2, 5

network fluids rather than simple liquids. A testable consequence is that, below c_{sat}, solutions should already contain a broad distribution of finite "pre-percolation" clusters (the χ of Example 2); such heterogeneous nano- and mesoscale clusters have been observed for phase-separating RNA-binding proteins [13].

4.2 Cytoskeletal and extracellular fibre networks

The cytoskeleton raises both questions of Example 4. For force transmission, Alvarado *et al.* reconstituted actin networks with tunable crosslink density and myosin activity [16]. Motors broke weakly crosslinked networks into separately contracting clusters, and the crossover from local to global contraction resembled connectivity percolation; motor activity tended to drive networks toward a critically connected state. Bueno *et al.* extended Flory–Stockmayer counting (Example 3) to actomyosin networks with several actin-binding proteins and estimated separate crosslinker concentrations for connectivity and for rigidity percolation [17]. For mechanics, collagen and fibrin networks are sub-isostatic ($z < 6$) and owe their rigidity to filament bending; their strong stiffening under strain is controlled by a strain-driven rigidity transition [9, 10].

4.3 Tissues and embryonic development

Treating cells as nodes and adhesive cell–cell contacts as links turns a tissue into a network. Petridou *et al.* applied rigidity percolation to the zebrafish blastoderm [18]. A small reduction in adhesion-dependent cell connectivity below a critical value triggered a sharp fluidization of the tissue, with viscosity dropping roughly tenfold within minutes; they predicted and verified hallmarks of a genuine transition, and showed that its spatial uniformity relies on synchronous cell divisions that change connectivity randomly, as percolation assumes. This is Example 4 carried out *in vivo*, with contacts per cell as the control parameter measured from images (Example 6).

4.4 Neurons, membranes, excitable media and epidemics

In cultured neural networks, Breskin *et al.* progressively weakened synapses chemically and followed the size of the largest group of neurons that responded together [19]. It shrank to zero at a critical coupling, as in bond percolation on a graph, and its shape revealed the network's degree distribution. In membranes, immobile proteins act as obstacles; near the obstacle percolation threshold the free area becomes a tenuous cluster and diffusion becomes anomalous [20], a standard reference point for

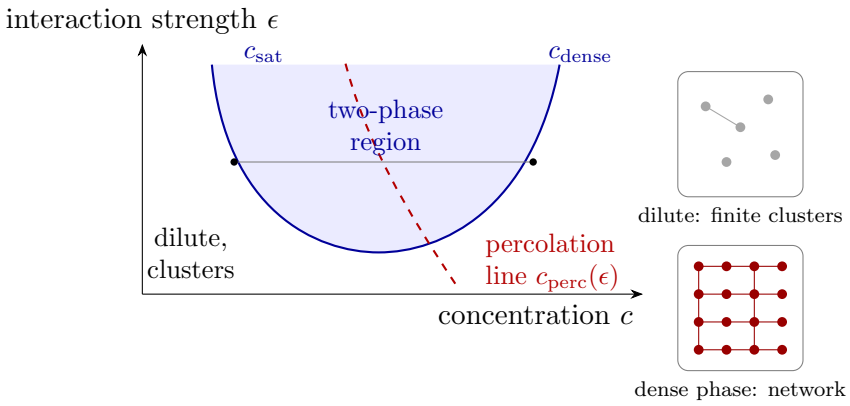


Figure 4: Schematic of phase separation coupled to percolation, after Refs. [11, 12]. Inside the coexistence region (shaded) a dilute phase at c_{sat} coexists with a dense phase at c_{dense}. When the percolation line (dashed) lies between them, the dense phase is a spanning network while the dilute phase holds only finite clusters.

single-particle tracking. Signal relay in excitable tissue (Example 5) and the final size of epidemics, which maps exactly onto bond percolation with transmissibility as p [3, 23], complete the list.

5 A practical checklist, and caveats

The examples suggest a recipe for asking whether percolation matters in a given system:

1. **Name the nodes and links** (molecules and bonds, filaments and crosslinks, cells and contacts).

2. **Pick a control parameter you can vary** experimentally: concentration, valence, crosslinker or adhesion-protein level, density, fraction of responsive cells.

3. **Decide what "spanning" must achieve.** Signal or material transport needs connectivity; mechanical stability needs rigidity (Example 4); a process moving forward in time (infection, relay with refractoriness) may need directed percolation.

4. **Estimate the threshold by counting**: branching ratio > 1 (Examples 2, 3) or Maxwell counting (Example 4). Expect the true threshold to be higher in two or three dimensions (Example 5).

5. **Check numerically** with realistic geometry (Example 6), and compare with data: a sharp rise of the largest cluster, a broad distribution of cluster sizes near threshold, and large sample-to-sample variability right at the transition.

Four caveats are specific to living matter. *Finite size*: an embryo may contain only 10^2–10^4 cells, so the transition is a smooth crossover [Fig. 2(c)]. *Correlations*: biological links are rarely independent (cooperative binding, patterned adhesion, mechanical feedback), which can shift the threshold or make the transition abrupt. *Reversibility*: bonds in condensates and cytoskeleta break and re-form, so a network that spans at one instant may flow over longer times; whether it behaves as gel or liquid depends on bond lifetimes. *Activity*: motors, divisions and ATP-driven remodelling keep systems out of equilibrium and can actively tune them toward or away from threshold [16].

6 Conclusion

Percolation asks when random local connections add up to a global connection. Most of its practical content can be obtained with simple tools: a branching-ratio argument gives thresholds on trees and in gelation theory, Maxwell counting separates connected from rigid networks, and a short simulation corrects both for real geometry. These ideas are now part of mainstream biophysics. They underlie the stickers-and-spacers theory of biomolecular condensates, the mechanics of cytoskeletal and extracellular networks, the fluidization of embryonic tissue, and the analysis of neuronal, excitable and epidemic networks. The open frontier is to extend them to correlated, reversible, finite and actively driven networks—the conditions that define living matter.

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